

Strategic Notes: Device Monitoring & Analytics Framework for LivingThings

Objective: Transform device data into reliable, operational decisions that optimize energy consumption, prevent failures, and enable organizational scale.

1. High-Level Architecture

The framework follows a five-stage pipeline:

Ingest → Store → Process → Analyze → Act

1.1 Data Ingestion

Ingest data into a centralized warehouse while performing basic sanity checks to maintain data integrity.

1.2 Data Storage

Store data and maintain historical records for up to 5 years to support auditing and long-term analysis requirements.

1.3 Data Processing

Normalize data and create multiple layers and pipelines through robust ETL (Extract, Transform, Load) processes to prepare data for downstream analysis.

1.4 Analytics & Serving

Utilize processed data for the creation of KPIs, dashboards, and analytics to track business metrics and operational performance.

1.5 Action Layer

Enable business stakeholders to take informed actions based on insights derived from analytics and dashboards.

2. Data Reliability & Governance

2.1 Schema Enforcement

- Enforce schema and contract validation at the gateway level
- Reject or flag malformed data at the point of entry

2.2 Data Quality Checks

1. **Ingestion SLAs:** Monitor and enforce service-level agreements for data ingestion
2. **Missing-Sequence Tracking:** Identify and track gaps in data sequences
3. **Late Data Watermarking:** Implement watermarks to handle delayed data arrivals
4. **Device Heartbeat Monitoring:** Expect and validate X messages per hour from each device

2.3 Data Retention Policy

- **Raw Telemetry:** Retain for 2–3 years
- **Aggregated Data:** Store hourly and daily aggregations for long-term trend analysis

2.4 Security & Privacy

1. Implement TLS encryption for data in transit
2. Enforce device authentication protocols
3. Apply least-privilege IAM (Identity and Access Management) policies
4. Avoid collection and storage of PII (Personally Identifiable Information)

2.5 Version Control

- Maintain immutable raw data storage
 - Implement versioned data transformations using tools like dbt (data build tool)
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3. Key Performance Indicators (KPIs)

3.1 Energy & Efficiency Metrics

1. **Energy per Area:** kWh/m²/day — enable cross-site comparisons
2. **Top 5 Sites by Energy Savings:** Track weekly, monthly, and quarterly performance
3. **Top 5 Sites with Energy Spikes:** Identify consumption anomalies weekly, monthly, and quarterly
4. **Top 10 Zero-Anomaly Devices:** Recognize most reliable devices
5. **Energy Savings by Segment:** Track weekly and monthly savings by device type and geography

3.2 Operational Metrics

1. **Top 5 Anomaly Categories:** Identify most common failure patterns
2. **Anomaly Rate:** Calculate number of anomalies per device-month
3. **Time-to-Detect (TTD):** Measure detection speed for anomalies
4. **Time-to-Remediate (TTR):** Track resolution time for identified issues
5. **False-Alert Rate:** Monitor percentage of false positive alerts

3.3 Business Metrics

1. **Energy Savings:** Track estimated vs. realized energy savings
 2. **Temperature Control SLA:** Percentage of sites meeting temperature control service-level agreements
 3. **Energy Consumption SLA:** Percentage of sites meeting energy consumption targets
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4. Dashboards (Role-Focused)

4.1 Executive Dashboard (One-Page View)

- Global energy consumption overview
- Trend analysis vs. baseline performance
- Top 10 sites by excess energy consumption
- Monthly savings summary (total, by geography, by device type)

4.2 Operations / Site Dashboard

- Interactive site map with device locations
- Real-time device status indicators
- Energy consumption by geography
- Recent anomalies and alerts
- Top faulty devices
- Top 5 high energy consumption devices by geography
- Top 5 reasons for device faults

4.3 Device Diagnostics Dashboard

- Per-device historical data (power consumption, temperatures, setpoint)
- Z-score timeline for anomaly detection
- Recent maintenance logs and service history

4.4 Analytics / Data Science Dashboard

- Correlation matrices for variable relationships
 - Cohort comparisons (by AC tonnage, star ratings, etc.)
 - Anomaly catalogue with root-cause tagging
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5. Team Structure & Process

5.1 Core Teams

5.1.1 Data Engineering

Responsible for data ingestion, pipeline development, and storage infrastructure.

5.1.2 Site Operations

Field technicians and escalation management for on-site issues.

5.1.3 Analytics / Insights

Development of energy optimization logic, anomaly detection algorithms, and dashboard creation.

5.1.4 Platform / SRE (Site Reliability Engineering)

System monitoring, alert management, and uptime maintenance.

5.2 Process Cadences

1. **Daily Check-ins:** Track progress updates and address blockers
2. **Monthly Analytics Sprint:** Deploy new KPIs and refine root-cause analysis rules
3. **Post-Mortem Reviews:** Conduct after P1 incidents to identify lessons learned
4. **Quarterly Reliability Review:** Comprehensive assessment of system performance

5.3 Ownership & Accountability

- Define clear ownership for each KPI, alert, and dashboard
 - Establish and maintain explicit SLAs for all deliverables
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6. Quick Wins (First 90 Days)

Phase 1: Foundation (Days 1-30)

1. Implement Ingestion SLA and Dashboard

- Establish baseline monitoring for data ingestion performance
- Create visibility dashboard for ingestion metrics

Phase 2: Detection (Days 31-60)

2. Deploy Z-Score Anomaly Detection

- Implement statistical anomaly detection algorithms
- Configure per-device alert mechanisms

Phase 3: Insights (Days 61-75)

3. Build Energy Performance Reporting

- Create "Top 10 sites by energy per area" daily report
- Conduct initial site audits based on findings

Phase 4: Action (Days 76-90)

4. Operationalize Incident Response

- Create one-page operations runbook for P1 incidents
 - Automate the most common remediation action
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Conclusion

This framework provides a comprehensive approach to device monitoring and analytics, enabling LivingThings to make data-driven decisions that optimize energy consumption, prevent operational failures, and support scalable growth. By following this structured implementation plan, the organization can achieve measurable improvements in efficiency and reliability within the first 90 days.